

Comparator construction can reverse the apparent advantage of peak-token MS/MS models over binned baselines

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Abstract

Comparisons between machine-learning models for tandem mass spectrometry (MS/MS) routinely place a model that reads individual peaks at full mass precision against a baseline built on binned intensity vectors, and read the difference as evidence about the representation. Such comparisons also change the encoder, the pooling rule, the acquisition information each arm receives and the optimisation each arm is given, so representation is confounded with comparator construction. We separate these sources in MS/MS substructure prediction on a public-source corpus of 414,992 spectra from GNPS, MassBank and MoNA spanning 66 instrument types, of which 372,200 are retained, evaluated on 10,058 scaffold-disjoint held-out structures under three seeds. An unconditioned binned baseline reaches 0.1130 macro AUPRC against a full-precision peak-token model’s 0.1709. Supplying the baseline with the acquisition covariates the peak-token arms already received raises it to 0.1418. An architecture-specific hyperparameter search, selected on the validation fold and never consulting the held-out fold, raises it to 0.2601 ± 0.0008 and reverses the ordering. The same search budget applied to the peak-token arm’s own capacity parameters raises it from 0.1567 to 0.1573 ± 0.0036 and selects the configuration that arm already had, leaving a tuned-against-tuned difference of -0.1028 . Because the optimised baseline carries $6.2\times$ the parameters, this does not establish that binned representations are superior; it establishes that the original ordering was not robust to comparator construction. Within a fixed peak-token encoder, by contrast, restoring full mass precision contributes $+0.0142$ ($+0.0097$ to $+0.0187$), the study’s one single-factor intervention. Representation claims therefore require matched comparator inputs and architecture-appropriate optimisation, and causal attribution should rest on within-architecture controls wherever possible.

Scientific contribution

We apply three methodological controls to one MS/MS substructure-prediction ladder. Metadata parity supplies every arm with the same acquisition covariates. Matched search effort gives each architecture the same search budget over its own capacity parameters. Within-encoder mass-axis quantisation moves the mass grid inside a fixed peak-token encoder. The first two controls test comparator construction, and together they reverse the apparent ordering between a full-precision peak-token model and a binned baseline. The third isolates a representation change, and yields $+0.0142$ macro AUPRC with no baseline entering the contrast. The three do not estimate comparable causal quantities and are not presented as doing so: only the third is a single-factor intervention. The contribution is methodological: a reported gain of this kind may not be robust to how the comparator is built and optimised.

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Keywords

Tandem mass spectrometry, structure annotation, spectral representation, mass defect, substructure prediction, ablation study, baseline design, experimental design

Introduction

Assigning a chemical structure to an MS/MS spectrum is a central challenge in untargeted metabolomics. Where the molecule appears in a reference library, the task can be approached by spectral matching, but many spectra acquired from biological extracts have no exact reference match [1]. Computational annotation therefore relates a spectrum to candidate structures through in-silico fragmentation and scoring, as in MetFrag [2]; prediction of structural fingerprints followed by database search, as in CSI:FingerID [3] and MIST [4]; or learned spectral representations, as in MS2DeepScore [5] and DreaMS [6].

For much of the recent machine-learning literature, a spectrum has been discretised onto a fixed grid of bins and presented to a model as a dense vector. This convention originated in spectral library search, where bins wider than the instrument’s mass error make matching robust to calibration drift. A 0.5 Da bin spans roughly 1,700 ppm at $m/z = 300$, whereas a modern time-of-flight instrument can operate at approximately 5 ppm. Such binning discards fine-scale information including mass defect, the difference between exact and nominal mass; for example, C_5H_7^+ (67.0542) and $\text{C}_4\text{H}_3\text{O}^+$ (67.0178) fall within the same 0.5 Da bin despite distinct elemental compositions.

Modern systems increasingly avoid coarse binning. CSI:FingerID scores fragmentation trees over exact masses, MIST tokenises peaks by chemical formula, DreaMS operates on peak lists, and TransExION embeds nominal mass and mass defect in separate channels [11]. Recent work also benchmarks alternative MS/MS featurisations [7] and examines how much mass resolution chemical-similarity prediction requires [8].

These two studies are the most direct empirical statements available about what mass precision is worth, and they point in different directions. Giné et al. [7] compare adaptive binning and spectrum hashing against learned embeddings across a large grid of models for structure annotation, and report that retrieval degrades markedly as the mass tolerance is relaxed. Working on chemical-similarity prediction rather than structure annotation, de Jonge et al. [8] adopt 0.1 Da bins and report that finer bins did not improve performance. The disagreement is itself a reason to ask the question in a form that admits a clean answer. Both establish that representation matters; neither isolates which part of a pipeline causes a particular gain, because a benchmark measures each featurisation bundled with the encoder that reads it and therefore estimates a joint effect at many points rather than separating it at one.

Two further strands bear directly on the present design. First, acquisition conditioning: collision energy, adduct, polarity and instrument determine which bonds break and which ions are observed, and models that receive this information as an explicit input are now common. Whether a comparator receives the same information is, as far as we are aware, not routinely reported. Second, multi-spectrum aggregation: a molecule is typically represented by several spectra acquired at different energies or in different adduct forms, and how those are combined is a modelling choice independent of how any single spectrum is encoded. Benchmarks such as MassSpecGym [9] have standardised evaluation and leakage control for this field, but the attribution question — which component of a pipeline a reported gain belongs to — is orthogonal to benchmarking and is not

resolved by it.

The reason attribution is hard here is structural. A dense binned vector and a sparse, variable-length peak set require different input interfaces, so moving from one to the other commonly changes the representation and the encoder together. Additional differences enter through pooling, training protocol and acquisition metadata. A two-arm comparison therefore estimates the joint difference between complete pipelines, not the causal contribution of the spectral representation alone.

We address this attribution problem in a controlled MS/MS substructure-prediction setting. Our key control quantises the mass axis *inside* the peak-token pipeline: peaks are snapped to the same 0.5 Da grid used by the dense baseline, collisions are merged, and the resulting tokens pass through the same peak-set encoder used for full-precision peaks. This makes the M1R-to-M1 contrast a direct test of mass-axis precision while holding the peak-token architecture fixed. We also correct an acquisition-metadata imbalance in the original dense baseline. Finally, we include an attention-free token sensitivity arm and a multi-spectrum aggregation arm to place the mass-axis effect in context. Throughout, we distinguish contrasts that are single-factor interventions from those that remain composite.

This paper does not propose a state-of-the-art structure-prediction architecture, and it does not benchmark external systems. CSI:FingerID, MIST, DreaMS, MS2DeepScore and TransExION differ from this work and from each other in training data, prediction target, objective and architecture; a number produced here is not comparable to a number produced there. What the paper offers is a control strategy — metadata parity and within-encoder mass quantisation — and an empirical demonstration of how much a reported representation gain can move when those controls are applied.

Methods

Corpora and splits

Both corpora are drawn from the CASMI 2026 training corpus, an aggregation of eleven public and in-house MS/MS libraries whose `ingest_lib` field records the source library.

The public-source primary corpus is the subset contributed by the GNPS, MassBank and MoNA libraries: 414,992 spectra over 52,428 structures, spanning 66 instrument types and 112 adducts. After filtering, 372,200 spectra over 50,287 structures and 24,627,256 peaks remain for modelling. We distinguish these source and retained counts throughout. All primary results are reported on this corpus.

The secondary corpus is a contiguous slice of two of the file’s twenty-one row groups, containing 262,144 spectra. Because the file is ordered by source library, this slice is not a random sample: it is 99.0% one in-house library and 1.0% another, and 99% of spectra come from a single instrument (timsTOF), across two instrument types and sixteen adducts. We use it as a deliberately contrasting sensitivity corpus. It is not an external validation set and not an independent replication: it is drawn from the same aggregate resource, processed by the same code, and is not redistributable.

Spectra were retained if precursor $m/z < 2,000$ and the reported precursor error was within 20 ppm. Peaks were retained above 0.1% of the base peak and below the precursor plus 1.5 Da, capped at the 256 most intense, with m/z held in double precision throughout preprocessing.

Molecular identity and the fold assignment

Molecular identity is the first block of the InChIKey, a 14-character hash of the connectivity layer

produced by standard InChI normalisation. That block excludes the stereochemical and isotopic layers, and normalisation fixes a reference protonation and tautomeric form before hashing, so records differing only in information outside the connectivity layer share one identity key and are assigned to a single fold. Records differing within that layer do not. The same identifier also merges records duplicated across source libraries, which is the case that matters most here: GNPS, MassBank and MoNA overlap, and a compound present in two of them must not be split between fitting and evaluation. Structures are additionally represented by a canonical SMILES string, used for scaffold computation and for reporting, but identity is defined by the InChIKey block alone.

Structures were then assigned to five folds by grouping generic Murcko scaffolds, with acyclic structures falling back to grouping by their own identifier, and fold sizes balanced greedily. Folds 2–4 were used for fitting, fold 1 for model selection and fold 0 for final evaluation. This design follows the generalisation-demanding practice used in MassSpecGym [9], while retaining source-library selection specific to the present study.

Scaffold grouping is stricter than a random split over spectra, and stricter than a split over structures, because it prevents a model from being evaluated on a molecule whose scaffold it has already seen. It is not equivalent to chemical-space independence. Two molecules with different generic Murcko scaffolds may still be close neighbours by any pharmacophoric or fragment-level measure, and BRICS fragments in particular recur across scaffolds by construction. A split defined on a graph edit distance between structures, as used by MassSpecGym [9], is stricter again; we did not adopt it here because our corpora are selected by source library rather than drawn from that benchmark.

Split-integrity audit

We verified the split rather than asserting it. On the primary corpus, across 50,287 structures and 372,200 retained spectra: no molecular identifier occurred in more than one fold (0 of 50,287); no scaffold group occurred in more than one fold (0 of 19,648 groups, of which 7,456 are acyclic structures keyed by their own identifier); every spectrum inherited the fold of its structure, with none unassigned. Fold sizes are 10,058, 10,058, 10,057, 10,057 and 10,057 structures. The same audit on the sensitivity corpus is likewise clean: 0 of 43,536 identifiers and 0 of 13,620 scaffold groups span folds.

Fold 0 of the primary corpus contains 10,058 structures and all 10,058 are evaluated. Fold 0 of the sensitivity corpus contains 8,708 structures, of which 16 are structures recoverable from the CASMI competition test set by exact spectral duplicate matching; those are excluded from both fitting and evaluation, leaving the 8,692 reported. The overlap was reported to the competition organisers, and the recovered structures were used only to remove those records from fitting and evaluation: no competition submission was derived from them. No such structures are present in the primary corpus, which is one further reason to treat it as primary. The audit is reproduced by `scripts/split_integrity_audit.py`.

Structural targets

Each structure was decomposed with BRICS and represented by the presence or absence of each of the 512 most frequent attachment-labelled fragments, with the vocabulary fitted on the training folds only. On the primary corpus the target matrix is 99.3% zero with a median of four positive targets per molecule. A total of 433 targets have at least five held-out positives, at a mean prevalence of 0.0085, which is the chance level for macro AUPRC. The sensitivity corpus is less sparse (97.1% zero, median 17 positive targets, 477 scored targets, mean prevalence 0.0093). Because the targets

are highly imbalanced, macro AUPRC is the primary metric; macro AUROC, micro AUPRC and Brier score are secondary summaries.

Five-arm ablation ladder

The five arms are ordered as M0, M1D, M1R, M1 and M2. The ladder decomposes the observed pipeline difference, but not every adjacent contrast is a single-factor intervention. M1R-to-M1 changes one component. M0-to-M1D, M1D-to-M1R and M2-to-M1 change several simultaneously and are labelled composite wherever they appear.

M0, metadata-matched dense baseline. Each spectrum is projected onto 2,048 fragment bins and 2,048 precursor-minus-fragment bins of width 0.5 Da over $[0, 1024)$, with square-root intensity accumulated within bins and the concatenated 4,096-dimensional vector L_2 -normalised. A two-layer perceptron encodes the vector. Collision energy, adduct, polarity and instrument embeddings are concatenated to the dense representation. This gives M0 access to the same acquisition variables as the peak-token arms, although the conditioning mechanism is not identical: the token encoder injects them through adaptive layer normalisation. We therefore describe the corrected comparator as metadata *matched*, not as mechanically identical in conditioning, and we treat the difference in mechanism as a limitation of the M1R-to-M0 contrast rather than a property of it.

M1R, rounded peak-token control. Each fragment mass m is assigned to a cell by $\lfloor m/w \rfloor$, with $w = 0.5$ Da, and moved to the cell centre. Peaks sharing a cell are merged by accumulating square-root intensity, matching the accumulator used by M0. The precursor is snapped to the same grid before the neutral loss is computed, preventing fine precursor mass from leaking back into the token representation through the subtraction $m^{\text{prec}} - m$. The resulting tokens pass through the same peak-token encoder used by M1. M1R and M1 have identical parameter counts (712,336) and differ only by whether mass quantisation is active. This is the principal representation control in the study.

M1D, attention-free token sensitivity arm. M1D uses the same rounded peak masses and acquisition variables as M1R, but removes self-attention from the token blocks. Because a learned pooling token cannot aggregate independent position-wise tokens in the absence of attention, M1D pools peaks by a masked mean. Removing attention also frees parameters, so its feed-forward layers were widened, yielding 714,528 parameters versus 712,336 for M1R. M1D therefore tests whether the attention-bearing configuration improves on an approximately capacity-matched attention-free token model. It does *not* isolate a self-attention effect, because pooling and feed-forward width change with it.

M1, full-precision peak-token model. M1 is identical to M1R except that mass quantisation is disabled, so the model receives the instrument-reported mass precision. The M1-to-M1R contrast isolates the mass-axis precision intervention within a fixed encoder, pooling rule, training protocol and parameter count.

M2, multi-spectrum aggregation. M2 is identical to M1 except that the mean over a molecule’s spectra is replaced by the measurement model specified below. The encoder, the mass axis, the peak pooling, the training protocol and the target heads are unchanged. The contrast is nonetheless composite rather than single-factor, and we label it so throughout: the measurement model replaces M1’s linear output head with a learned per-target prior, covariate-dependent scaling and a logistic-Gaussian output map, all of which act even on a molecule represented by a single spectrum, where there is nothing to aggregate. Sect. 7 reports the stratification that separates the two components empirically.

Peak encoding

Each peak is encoded as a token. For a peak at $m/z = m$ in a spectrum with precursor mass m^{prec} , we form a multi-scale sinusoidal embedding

$$\varphi_L(m) = [\sin(2\pi m/\lambda_1), \cos(2\pi m/\lambda_1), \dots, \sin(2\pi m/\lambda_L), \cos(2\pi m/\lambda_L)]^\top, \quad (1)$$

with $L = 32$ wavelengths spaced geometrically from 2 mDa to 10^3 Da, applied to both fragment mass and neutral loss. The token also receives the explicit defect $\delta = m - \text{round}(m)$, square-root and log-transformed intensity, and intensity rank. At a wavelength of 2 mDa and $m/z = 342$ the phase exceeds 10^6 , so the phase is evaluated in double precision and reduced modulo 2π before casting to single precision. Tokens are encoded by a four-block Transformer of width 128 with eight heads, with acquisition covariates entering each block through adaptive layer normalisation.

Acquisition covariates

For spectrum k of molecule i , the covariate vector $c_{ik} \in \mathbb{R}^{32}$ is produced by a small encoder from learned embeddings of adduct, polarity and instrument family, a six-knot spline basis in harmonised collision energy, the log precursor mass, and an explicit indicator for missing collision energy. Energies reported as normalised collision energy are converted to eV by $\text{eV} = \text{NCE} \cdot m/z / 500$; energies in unrecognised units are treated as missing rather than placed on the eV scale. Missing metadata are therefore represented explicitly rather than imputed, and the same c_{ik} is supplied to every arm.

Multi-spectrum aggregation in M2

Let molecule i have spectra $x_{i1}, \dots, x_{iK_i}$ with encoder outputs $h_{ik} \in \mathbb{R}^{128}$ and covariate vectors c_{ik} . M2 aggregates at two levels, and we state both because the second is where the arm’s parameters lie.

Acquisition-dependent reliability. For each target $j \in \{1, \dots, 512\}$ the covariates produce three quantities by rank-one affine maps of c_{ik} ,

$$\alpha_j(c_{ik}) = \alpha_j^0 + a_j^\top c_{ik}, \quad (2)$$

$$\beta_j(c_{ik}) = \text{softplus}(\beta_j^0 + b_j^\top c_{ik}) + \varepsilon, \quad \varepsilon = 10^{-2}, \quad (3)$$

$$\log \sigma_j(c_{ik}) = s_j^0 + s_j^\top c_{ik}, \quad \text{clamped to } [-4, 4], \quad (4)$$

an offset, a sensitivity and a noise scale. Positivity of β_j is enforced by the softplus and the floor ε ; positivity of σ_j by the exponential, and the clamp bounds it away from zero and infinity. The per-spectrum precision for target j is

$$w_{ikj} = \beta_j(c_{ik})^2 / \sigma_j(c_{ik})^2, \quad (5)$$

clamped above at 10^3 . Note that w_{ikj} depends on the acquisition covariates only, not on h_{ik} , so it can be evaluated before the spectra are pooled.

Representation-level pooling. The molecule representation is a normalised weighted mean over the observed spectra, using the precision of Eq. (5) averaged over targets,

$$\bar{w}_{ik} = \frac{1}{512} \sum_j w_{ikj}, \quad h_i = \frac{\sum_{k=1}^{K_i} \bar{w}_{ik} h_{ik}}{\sum_{l=1}^{K_i} \bar{w}_{il}}. \quad (6)$$

The weights are non-negative and, after the division, sum to one; masked positions contribute zero.

M1 and the other arms use the same expression with $\bar{w}_{ik} \equiv 1$, that is, a masked mean.

Evidence-level pooling. A per-spectrum evidence logit $\ell_{ikj} = (Wh_{ik} + b)_j$ is combined across spectra by Gaussian-conjugate pooling. With the prior mean $\mu_{ij} = (W'h_i + b')_j$ from Eq. (6) and a learned per-target prior precision τ_j^{-2} ,

$$S_{ij}^{(1)} = \sum_{k=1}^{K_i} \frac{\beta_j(c_{ik})(\ell_{ikj} - \alpha_j(c_{ik}))}{\sigma_j(c_{ik})^2}, \quad S_{ij}^{(2)} = \sum_{k=1}^{K_i} w_{ikj}, \quad (7)$$

$$v_{ij} = (\tau_j^{-2} + S_{ij}^{(2)})^{-1}, \quad m_{ij} = v_{ij}(\mu_{ij}\tau_j^{-2} + S_{ij}^{(1)}), \quad (8)$$

and the reported probability uses the logistic-Gaussian approximation

$$\text{logit } \hat{p}_{ij} = m_{ij} / \sqrt{1 + \frac{\pi}{8} v_{ij}}. \quad (9)$$

Because $S^{(1)}$ and $S^{(2)}$ are sums over spectra, a molecule’s spectra need not be processed in one pass: statistics accumulated over chunks give the same posterior, so no cap on K_i is required. All aggregation parameters ($\alpha^0, a, \beta^0, b, s^0, s, \tau, W, b, W', b'$) are trained jointly with the encoder by the same objective; none is fitted post hoc.

Bin-width sweep

To examine the precision scale continuously, we repeated the M1R construction at $w = 0.25, 0.1, 0.05$ and 0.01 Da, holding the encoder, parameter count, training schedule and seed fixed. Together with $w = 0.5$ and full precision, this produces six mass-resolution settings. The sweep was run on the sensitivity corpus only and each intermediate width was trained once; it is exploratory except at its endpoints.

Training protocol

Every arm was trained for 15 epochs with AdamW, a one-cycle learning-rate schedule peaking at 10^{-3} , batches of 16 molecules, mixed precision and gradient clipping at 1.0. The objective was binary cross-entropy over the 512 BRICS-presence targets plus an auxiliary negative-binomial term on 88 functional-group counts. Checkpoints were selected by validation macro AUPRC on fold 1. The five ladder arms on the primary corpus and four arms on the sensitivity corpus were each trained under seeds $\{2026, 7, 13\}$, for 27 ladder runs. The two searches add 36 validation-only selection runs and six further runs for the two retrained arms, and the bin-width sweep adds four single-seed runs.

All arms were implemented for this study in PyTorch 2.11.0 (CUDA 12.8) under Python 3.14, with NumPy 2.4.4, pandas 2.3.3 and RDKit 2026.3.6, and trained on a single NVIDIA RTX 3070. M0 reproduces the binned convention described in the Introduction rather than any particular published system, and no external implementation was used as a comparator; every difference reported here is between arms of one codebase, which is what makes the within-architecture controls necessary rather than optional.

Hyperparameter selection, and the searches that make it symmetric

No hyperparameter search was performed for any arm. This is not a symmetric choice and we state it plainly: the schedule above was selected for the peak-token Transformer and inherited unchanged by the dense baseline, so M0 and the token arms are not equidistant from their respective optima. Because the central claim of this paper concerns comparator fairness, a comparator that is merely untuned is a weaker instrument than the argument requires. The metadata correction reported

below, worth more than any contrast in the study, is direct evidence that M0 was not near its ceiling.

We therefore prespecify a modest tuning experiment for the dense baseline. The scaffold split is fixed and unchanged; folds 2–4 are used for fitting and fold 1 for model selection; fold 0 is not consulted at any point during selection. The grid varies learning rate $\in \{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, hidden width $\in \{256, 512, 1024\}$ and dropout $\in \{0.1, 0.3\}$, all other settings held at the values above. The configuration with the best fold-1 macro AUPRC is then trained under the three seeds and evaluated once on fold 0. The purpose is to test whether the residual M1R-to-M0 difference survives a stronger dense comparator; if the tuned baseline changes the decomposition, every contrast involving M0 must be recomputed and reported against the tuned baseline, and the contrasts not involving M0 — M1-to-M1R and M2-to-M1 — are unaffected by construction. The design is stated here in the prospective form in which it was fixed, before the outcome was known.

Selection runs evaluate the validation fold only. The driver refuses to proceed if a selection run has produced a held-out score, so the search cannot be contaminated by fold 0 even accidentally. As a check that the grid does what it claims, it reproduces the incumbent configuration’s validation score exactly at the incumbent’s own settings.

A search on one side of a contrast and not the other reverses an unfairness rather than removing it, so the same grid was then run on the peak-token side. The width axis maps to the knob that carries capacity in that arm — the feed-forward width of the peak-token blocks — with d_{model} held at 128 so that the representation width is unchanged and the search varies capacity rather than the interface. Learning rate and dropout ranges are identical, selection is again on fold 1, and fold 0 is again not consulted. We call the resulting arm M1RT, and the M1RT-to-M0T contrast is the tuned-against-tuned comparison: both sides searched over the same grid, on the same folds, with the held-out fold consulted by neither. It is the comparison on which Sect. 2 rests. The design equalises search effort over architecture-specific capacity parameters; it does not equalise parameter count or effective capacity, and we do not treat it as doing so.

An exact attention ablation, which this study specifies and does not run

The M1D-to-M1R contrast reported below does not price self-attention, and the reason is worth stating because it is a property of the comparison rather than of the measurement. Removing attention forces a second change. M1R pools peaks with a learned token, which is a query that aggregates by attending to them, so an arm without attention cannot use one and M1D pools by a masked mean instead. Removing attention also frees parameters, which were returned as feed-forward width. Three things therefore differ between the two arms at once, and a single measurement cannot be divided among them.

The consequence is not that the estimate is imprecise but that it is uninterpretable in a particular direction: if self-attention helped and masked-mean pooling helped by a similar amount, the two would cancel and the contrast would read as approximately zero — which is close to what it does read. An interval containing zero is therefore consistent with attention mattering and with attention not mattering, and the design cannot separate the two.

The experiment that would price the operator holds the input, the feed-forward width, the training protocol and the parameter count fixed as closely as the operator allows, and toggles self-attention alone. Its essential feature is that *both* arms pool peaks by a masked mean, so that the pooling rule no longer changes with the operator; the attention-bearing arm gives up its learned token to buy that. No quantity in this paper prices self-attention.

OUTSTANDING EXPERIMENT. Run the exact attention ablation and let it replace the M1D-to-M1R contrast of Sect. 6, which prices a package rather than an operator.

Evaluation and three sources of uncertainty

Macro AUPRC is reported over targets with at least five held-out positives. Three distinct sources of variability bear on the contrasts, and we report them separately rather than combining them into one interval.

(i) *Seed variability.* Each arm is trained three times under different initialisations. We report the standard deviation of each contrast across seeds, and we check every claimed ordering within each seed rather than reading it off the means.

(ii) *Target-resampling uncertainty.* For paired contrasts, each target’s average precision is first averaged over the three seeds; per-target differences are then computed and their mean is resampled over targets with 5,000 bootstrap draws. We call the resulting intervals *target-resampling intervals*. They describe heterogeneity across the scored target set at a fixed held-out set of molecules. They do *not* describe uncertainty arising from which molecules were held out, and BRICS targets are not statistically independent because fragments co-occur, so these intervals are narrower than a fully independent resampling would give.

(iii) *Test-set uncertainty over structures.* To quantify the sampling of held-out molecules we add a *scaffold-cluster bootstrap*. Scaffold groups, not molecules or spectra, are the resampling unit, because molecules sharing a generic Murcko scaffold were assigned to a fold together and are not exchangeable with one another. Whole scaffold groups are drawn with replacement, the full macro AUPRC is recomputed per arm within each resample over the scored targets, and the contrasts are formed inside the resample so that pairing is preserved. We use 2,000 draws over the 3,784 scaffold clusters that partition the 10,058 held-out structures.

The two resamplings vary different things and are not interchangeable. The target-resampling bootstrap holds the held-out structures fixed and varies the target set; the scaffold-cluster bootstrap holds the target set fixed and varies the structural composition of the held-out set; seed spread varies initialisation alone.

Average precision is not a linear functional of the evaluation set. Resampling with replacement duplicates structures, and duplication changes the ranking denominators from which precision is computed, so in general $\mathbb{E}_{\text{boot}}[\hat{\Delta}^*] \neq \hat{\Delta}$ even when the resampling scheme is correct. A scaffold-cluster bootstrap mean that sits away from its point estimate is therefore expected behaviour rather than evidence of an error, and we report both quantities side by side so that the displacement can be read directly. We use the percentile interval without bias correction and interpret all three sources descriptively. Neither interval is a confidence interval for the population of all possible MS/MS tasks.

Results

Each contrast below is labelled ONE-FACTOR INTERVENTION where the arms differ in exactly one component, and COMPOSITE CONTRAST where they differ in more than one. Causal language is used only for the former.

1. Metadata matching substantially reduces the apparent two-arm gap

The initial dense baseline did not use collision energy, adduct, polarity or instrument variables, whereas the peak-token models did. Adding the same acquisition variables to M0 by concatenating their embedding to the dense spectral vector raises M0 from 0.1130 to 0.1418 macro AUPRC,

an improvement of +0.0288. Consequently the apparent M1-to-M0 advantage falls from +0.0579 to +0.0291. About half of the original two-arm difference was attributable to unequal metadata availability rather than to mass precision or peak-token modelling.

At this point in the study the metadata correction was the largest single quantity measured, and it already belonged to the construction of the comparator rather than to any modelling choice under test. Sect. 2 reports a second intervention on the comparator that is four times larger again. Because M0 and the peak-token arms inject the metadata through different mechanisms, we describe the corrected comparator as metadata matched, not as mechanically identical in conditioning. Table 1 summarises the five arms under the corrected baseline, and Fig. 1 places that correction beside the two that follow.

Sects. 2 and 3 take the comparator further, and are reported next because the comparator corrections are larger than any modelling effect that follows them. The contrasts in Sects. 4 onwards are measured against this metadata-matched but *untuned* baseline, which is the comparator a conventional study would use; those that involve M0 are recomputed against the tuned baseline in Table 2, and those that do not are unaffected by construction.

Table 1: Five-arm ablation ladder on the primary corpus, measured against the metadata-matched but *untuned* dense baseline. Sect. 2 reports the same arms against the tuned one, where the ordering reverses. The AUPRC column is macro AUPRC on 10,058 held-out molecules, averaged over three seeds, and SD its spread over those seeds; “attn.” is self-attention. Chance is 0.0085 over 433 scored targets. Pooling is given on two axes because different pairs of arms differ on different ones: peak pooling within a spectrum distinguishes M1D from M1R, spectrum pooling within a molecule — the “Spectra” column, where “meas. model” is the measurement model of Eqs. (2)–(9) — distinguishes M2 from M1.

Arm	Mass axis	Encoder	Peak pooling	Spectra	Parameters	AUPRC	SD
M0 (untuned)	0.5 Da bins	dense MLP	bin sum	mean	1,172,752	0.1418	0.0013
M1D	0.5 Da grid	tokens, no attn.	masked mean	mean	714,528	0.1590	0.0024
M1R	0.5 Da grid	tokens, attn.	learned token	mean	712,336	0.1567	0.0025
M1	full precision	tokens, attn.	learned token	mean	712,336	0.1709	0.0022
M2	full precision	tokens, attn.	learned token	meas. model	829,584	0.1888	0.0015

2. Architecture-specific tuning of the comparator reverses the ordering

The search prespecified in Methods was run: 18 configurations, fitted on folds 2–4, selected on fold 1, fold 0 never consulted. The grid reproduced the incumbent’s validation score exactly at the incumbent’s settings (0.1702), and selected $\text{lr} = 3 \times 10^{-3}$, hidden width 1024, dropout 0.3 at a validation score of 0.2933. Retrained under the three seeds and evaluated once on fold 0, the tuned baseline reaches 0.2601 ± 0.0008 against the untuned 0.1418 (Table 2). Tuning the comparator was worth +0.1183 (+0.1051 to +0.1312, improving 87.3% of targets), four times the metadata correction and larger than every modelling effect in this study combined.

3. Matched search effort on the peak-token arm does not restore the ordering

The same grid was then run on the peak-token side, with the width axis mapped to the feed-forward width as specified in Methods: again 18 configurations, again selected on fold 1, again with fold 0 never consulted. It selected $\text{lr} = 10^{-3}$, feed-forward width 256, dropout 0.1 at a validation score of 0.2026 — which is M1R’s incumbent configuration, the settings the arm already had. Retrained under the three seeds, the tuned arm M1RT reaches 0.1573 ± 0.0036 against the untuned 0.1567 ± 0.0025 . The matched search was therefore worth +0.0006: an interval of -0.0017

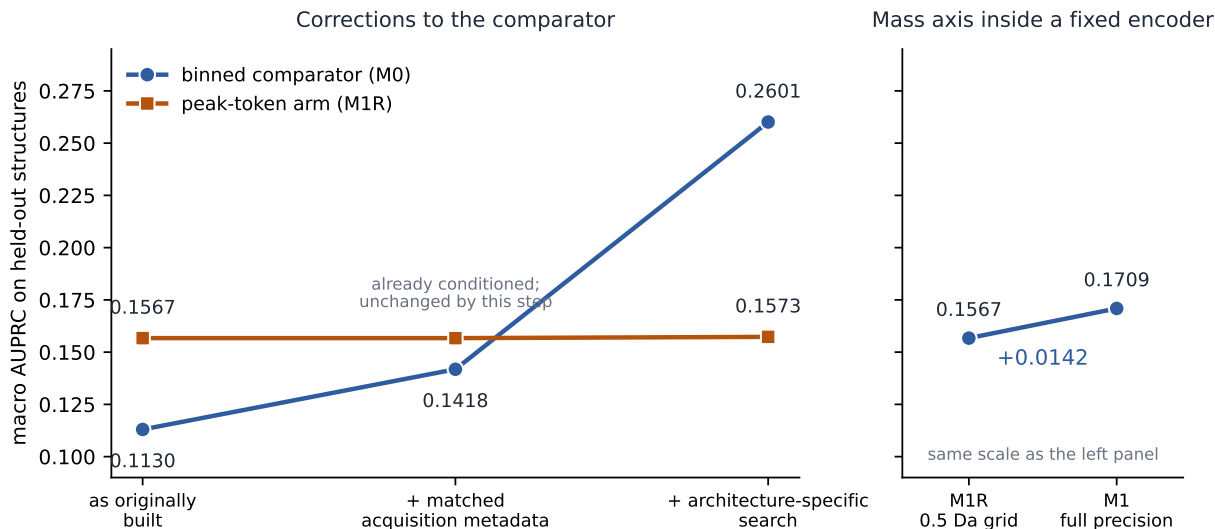


Figure 1: Left: the binned comparator under two corrections that never touch the treatment, against the peak-token arm receiving the same search budget. The peak-token arm is unchanged by the metadata step because it already received those covariates. Right: the one contrast in which a single component moves and no comparator enters, quantising the mass axis inside a fixed peak-token encoder. All values are three-seed means on the 10,058 held-out structures; both panels share one scale.

to +0.0029 over targets and -0.0019 to $+0.0041$ over scaffold clusters, 52.2% of targets improved, and the sign changing across seeds. Nothing measurable.

The tuned-against-tuned comparison is accordingly where the untuned one already was. M1RT minus M0T is -0.1028 (-0.1164 to -0.0898 over targets, -0.1088 to -0.0874 over scaffold clusters), negative in all three seeds, with the peak-token arm ahead on 21.9% of scored targets. The remaining arms sit below the tuned baseline by -0.1011 , -0.0892 and -0.0713 , every interval excluding zero (Table 2). Both sides have now received the same search budget applied to architecture-specific hyperparameters, on the same folds, with the held-out fold consulted by neither, and the ordering does not move. The experiment equalises search effort, not parameter count or effective model capacity: the dense grid spans 1,172,752 to 4,442,128 parameters while the peak-token grid spans 712,336 to 1,501,840, so the comparator’s grid reaches roughly three times the capacity the treatment’s does.

Three things must be said about what this does and does not show.

It is still not evidence that binned representations are superior. The tuned baseline carries $6.2\times$ M1RT’s parameters, so the comparison is not capacity matched, and the grid is modest — six learning-rate-by-width combinations at two dropout rates — with the baseline’s selection on its boundary in both axes. What can no longer be said is that the ordering reflects unequal search effort, because the effort was equalised and the ordering did not move.

The mechanism is underfitting, not regularisation. The untuned baseline ends training at a presence loss of 0.01885 against the tuned arm’s 0.01084: the tuned model fits the training data better *and* generalises better, with a smaller relative validation-to-test gap (11.6% against 15.8%). Its validation curve was still rising at the final epoch under the shared schedule.

The cause is traceable to the design, which is the part that generalises beyond this codebase. The binned encoder’s hidden width was not chosen for a 4,096-dimensional input; it was set to

Table 2: Both arms after their own search over the same grid, and the contrasts that follow. Selection used fold 1 only, on both sides. The dense baseline’s selected configuration is $\text{lr } 3 \times 10^{-3}$, width 1024, dropout 0.3, on the boundary of the grid in both axes, which indicates that the search may not have located that architecture’s optimum, and the direction of any change on the held-out fold under a broader search cannot be assumed; the peak-token arm’s selected configuration is $\text{lr } 10^{-3}$, feed-forward width 256, dropout 0.1, which is the configuration it already had. Intervals are 95% target-resampling and scaffold-cluster intervals, computed from the saved held-out predictions by the procedures of Table 3 and Table 6.

Arm	Parameters	Macro AUPRC	Seed SD
M0, untuned	1,172,752	0.1418	0.0013
M0T, tuned dense baseline	4,442,128	0.2601	0.0008
M1R, untuned	712,336	0.1567	0.0025
M1RT, tuned peak-token arm	712,336	0.1573	0.0036
M1D	714,528	0.1590	0.0024
M1	712,336	0.1709	0.0022
M2	829,584	0.1888	0.0015

Contrast	Isolates	Δ AUPRC	95% target	95% scaffold
M0T – M0	tuning the comparator	+0.1183	[+0.1051, +0.1312]	[+0.1099, +0.1351]
M1RT – M1R	tuning the treatment	+0.0006	[−0.0017, +0.0029]	[−0.0019, +0.0041]
M1RT – M0T	tuned against tuned	−0.1028	[−0.1164, −0.0898]	[−0.1088, −0.0874]
M1R – M0T	untuned treatment	−0.1033	[−0.1167, −0.0901]	[−0.1089, −0.0890]
M1D – M0T	untuned treatment	−0.1011	[−0.1138, −0.0881]	[−0.1098, −0.0945]
M1 – M0T	untuned treatment	−0.0892	[−0.1028, −0.0760]	[−0.0953, −0.0717]
M2 – M0T	untuned treatment	−0.0713	[−0.0848, −0.0584]	[−0.0779, −0.0513]

twice d_{model} , the peak-token encoder’s width. The comparator’s only capacity parameter was a function of the treatment’s hyperparameter, and the learning-rate schedule was likewise selected for the treatment’s optimisation behaviour. A binned-versus-peak comparison can therefore be biased before any data is seen, by inheritance rather than by intent, and this is checkable in any codebase in minutes.

The two searches make that inheritance visible rather than inferred. The rate the dense baseline selected, 3×10^{-3} , is the worst region of the grid for the peak-token arm: all six configurations at that rate score between 0.033 and 0.058 on validation, against 0.2026 at 10^{-3} . The width axis says the same in reverse — the dense baseline selects the largest width on the grid and the peak-token arm the smallest. A single shared schedule cannot be neutral between two arms whose optima sit at opposite corners of it, and the one this study inherited was the treatment’s.

4. Restoring mass precision within a fixed encoder gives a reproducible gain (ONE-FACTOR INTERVENTION)

Against the metadata-matched baseline, M1 reaches 0.1709, a paired target-level gain of +0.0291 (95% target-resampling interval +0.0208 to +0.0371), with 72.5% of targets improved. The rounding control divides that numerical difference into +0.0149 for M1R minus M0 and +0.0142 for M1 minus M1R.

M1R and M1 are identical in encoder, pooling, parameter count, schedule and seed set, and differ only in whether the mass axis is quantised to the baseline’s grid. This is the study’s one single-factor intervention. The gain is +0.0142 (+0.0097 to +0.0187), improving 64.9% of targets, and it is the most reproducible contrast across seeds in the study, with a seed standard deviation of 0.0004

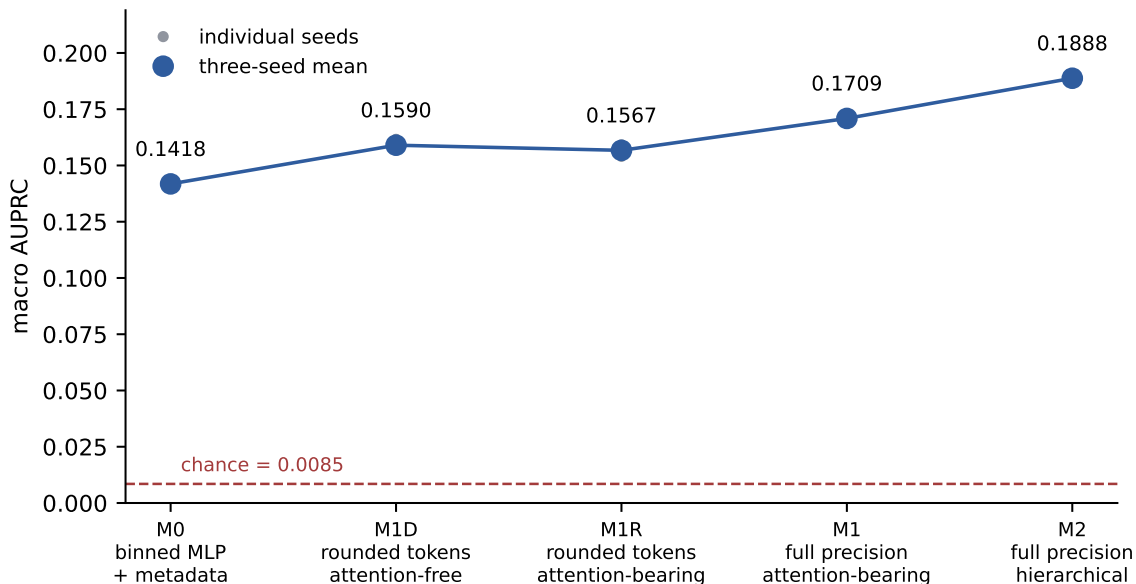


Figure 2: Held-out macro AUPRC for the five arms on the primary corpus. Large markers are three-seed means, small markers the individual seeds. The seeds are shown because two of the increments compared below cannot be ordered: their difference changes sign across these runs. The searched arms M0T and M1RT of Sect. 2 are not shown here; Table 2 gives them.

against 0.0013 for M1R minus M0 and 0.0009 for aggregation. Mass precision therefore carries predictive information that the binned representation removes, and the size of that information is measured here without confounding by architecture.

5. The rounded token-encoder package remains a composite comparison (COMPOSITE CONTRAST)

The M1R-to-M0 difference of +0.0149 (+0.0080 to +0.0221; 66.7% of targets) is broader than an encoder change. The two arms differ in input interface (dense vector against per-peak tokens), in network family (perceptron against Transformer), in peak pooling, and in the mechanism by which acquisition covariates are injected (concatenation against adaptive layer normalisation). We therefore call it the *rounded token-encoder package* difference and do not attribute it to any one of those components.

Its numerical similarity to the mass-axis increment supports a negative statement rather than a positive one: after metadata matching, the residual advantage of the full-precision peak-token model is not dominated by mass resolution. It does not establish that architecture and mass precision have equal causal effects, and the two cannot be ordered reliably — their difference is +0.0019, +0.0015 and −0.0012 across the three seeds, changing sign.

6. The attention sensitivity analysis does not identify an attention effect (COMPOSITE CONTRAST)

M1D minus M0 is +0.0172 (+0.0102 to +0.0242), and M1R minus M1D is −0.0023 (−0.0066 to +0.0020), an interval containing zero, with 54.3% of targets improved and a positive mean in only one seed of three. The attention-bearing rounded-token model is therefore not shown to outperform the approximately capacity-matched attention-free one in this comparison.

We do not read the negative mean as evidence that attention is harmful, and we do not read

Table 3: Paired target-level contrasts on the primary corpus, with 95% target-resampling intervals after averaging target AP across the three seeds. Every row is measured against the untuned dense baseline of Table 1; Table 2 gives the same arms against the tuned one. The Kind column states what each contrast changes; causal language in the text is confined to the single-factor row.

Contrast	Interpretation	Kind	Δ AUPRC	95% interval	Targets improved
M1D – M0	attention-free rounded token-encoder package	composite	+0.0172	[+0.0102, +0.0242]	58.0%
M1R – M1D	attention-bearing vs approximately capacity-matched attention-free configuration	composite	-0.0023	[-0.0066, +0.0020]	54.3%
M1R – M0	rounded token-encoder package	composite	+0.0149	[+0.0080, +0.0221]	66.7%
M1 – M1R	mass-axis precision	one-factor	+0.0142	[+0.0097, +0.0187]	64.9%
M2 – M1	measurement model: aggregation and calibration	composite	+0.0179	[+0.0136, +0.0223]	63.7%
M1 – M0	joint token-pipeline and mass-axis difference	composite	+0.0291	[+0.0208, +0.0371]	72.5%
M2 – M0	all changes combined	composite	+0.0470	[+0.0377, +0.0566]	73.9%

the result as an attention effect at all: M1D and M1R differ in the attention operator, in peak pooling and in feed-forward width together, because an attention-free arm cannot use a learned pooling token. An interval containing zero is what two offsetting effects would also produce, so it is evidence about the package and not about the operator. We therefore describe the comparison as attention-bearing versus approximately capacity-matched attention-free configuration, and treat it as a sensitivity analysis. The defensible statement is narrow: the tested attention-bearing configuration was not necessary to reach the rounded-token performance observed here. The experiment that would identify the operator’s contribution is specified in Methods and remains future work.

7. The measurement model improves prediction, and aggregation is only one component (COMPOSITE CONTRAST)

Replacing the masked mean over a molecule’s spectra with the measurement model of Eqs. (2)–(9) is worth +0.0179 (+0.0136 to +0.0223; 63.7% of targets).

This is not a pure aggregation effect. Stratifying the held-out set by the number of spectra per molecule gives +0.0085 on the 3,443 molecules represented by a *single* spectrum, +0.0101 on the 2,137 with two spectra and +0.0143 on the 4,478 with three or four. The gain among single-spectrum molecules demonstrates that a substantial component of M2’s advantage arises from changes in the output and evidence model that operate even when multi-spectrum aggregation is impossible: at $K_i = 1$, Eqs. (7)–(9) still apply a learned per-target prior, covariate-dependent scaling and the logistic–Gaussian output map, none of which M1’s linear head has.

The increasing gain with the number of spectra is consistent with an additional contribution from aggregation, but does not isolate that contribution causally. The three strata are observationally different populations of molecules, not randomised assignments of spectrum count, so the differences between them confound the number of spectra with whatever else distinguishes molecules measured once from molecules measured repeatedly. M2 – M1 is reported throughout as a composite contrast, and no split of the +0.0179 into calibration and aggregation components is claimed.

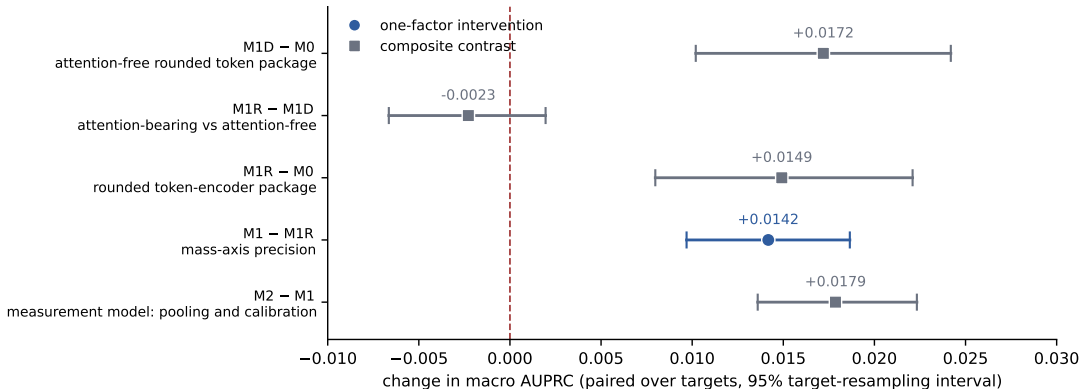


Figure 3: Paired target-level contrasts on the primary corpus with 95% target-resampling intervals, all measured against the untuned dense baseline. Marker shape as well as colour separates the one single-factor intervention, $M1 - M1R$, from the composite contrasts, whose arms differ in more than the named component. Every contrast here is superseded by Sect. 2 wherever $M0$ enters it.

The monotone trend across strata is consistent with a genuine aggregation component on top of that, and it carries a second implication: the evaluation uses at most four spectra per molecule, while 2,755 of the 10,058 held-out molecules have more available. The reported figure is therefore measured under a cap that Eqs. (7)–(8) do not require, and probably understates what the pooling contributes.

It exceeds the $M1R$ -to- $M0$ package in all three seeds by +0.0009, +0.0030 and +0.0049, and the mass-axis intervention in all three by +0.0028, +0.0045 and +0.0037; the same direction against the package holds in all three seeds on the sensitivity corpus. With three seeds per corpus we treat this as a consistent empirical ordering in these runs rather than an established general ranking. Its relevance to the paper’s argument is that a component with nothing to do with spectral representation produces an effect of the same magnitude as the representation change being studied.

8. A contrasting acquisition regime shows qualitatively similar patterns

The same ladder was evaluated on the near-homogeneous sensitivity corpus (Table 4). All three principal increments are larger there, but the broad pattern is similar: the mass-axis and rounded token-encoder increments are of comparable size, their difference again changes sign across seeds (−0.0016, +0.0006, −0.0090), and aggregation is again substantial. The corresponding share of the $M1$ -to- $M0$ difference assigned numerically to $M1R$ -to- $M0$ is 46.7% there against 51.3% here.

This is a sensitivity analysis, not an independent replication: the corpus is a contiguous in-house slice of the same aggregate CASMI resource, processed by the same code, and is not redistributable.

Table 4: Contrasts on the near-homogeneous sensitivity corpus and the primary corpus. “Token-encoder package” denotes $M1R$ – $M0$ and is composite. Absorption is measured on each corpus’s own held-out spectra.

Contrast	Sensitivity corpus	SD	Primary corpus	SD
Token-encoder package ($M1R$ – $M0$), composite	+0.0234	0.0002	+0.0149	0.0013
Mass axis ($M1$ – $M1R$), one-factor	+0.0268	0.0049	+0.0142	0.0004
Measurement model ($M2$ – $M1$), composite	+0.0322	0.0035	+0.0179	0.0009
Share of $M1$ – $M0$ assigned numerically to $M1R$ – $M0$	46.7%	–	51.3%	–
Peaks absorbed at 0.5 Da in held-out spectra	34.4%	–	23.6%	–

The metadata correction also appears on both corpora: supplying acquisition covariates to M0 improves macro AUPRC by +0.0288 on the primary corpus and +0.0142 on the sensitivity corpus. The larger gain in the more heterogeneous corpus is consistent with acquisition context becoming more valuable as experimental conditions diversify, but with two corpora this is an observation, not a tested relation.

9. Peak absorption describes the information coarse binning removes

On the primary corpus’s held-out spectra, 23.6% of the 3,797,166 peaks fall into a 0.5 Da bin already occupied by another peak from the same spectrum and are therefore absorbed. The corresponding fraction on the sensitivity corpus’s held-out spectra is 34.4%. At 0.01 Da the reported held-out absorption fractions fall to 2.9% and 6.5% respectively. Absorption is reported throughout on each corpus’s own held-out spectra; the same quantity computed over all retained spectra is 37.5% rather than 34.4% on the sensitivity corpus, and we quote the held-out figure because it matches the set on which the effects are measured.

A merged peak also loses compositional discrimination: enumerating plausible CHNOS compositions within 10 ppm places 328 candidate fragment formulae inside a single 0.5 Da bin at $m/z = 300$, consistent with the dependence of formula ambiguity on mass tolerance described by Kind and Fiehn [10].

The corpus that loses more peaks at 0.5 Da is also the corpus in which the mass-axis increment is larger (34.4% and +0.0268 versus 23.6% and +0.0142). This association is descriptive and does not identify a mechanism; with two corpora it cannot. One plausible explanation is that the sensitivity corpus is almost entirely high-resolution timsTOF data whereas the primary corpus spans instruments of varying resolution, but that remains a hypothesis until the mass-axis increment is stratified directly by instrument resolution.

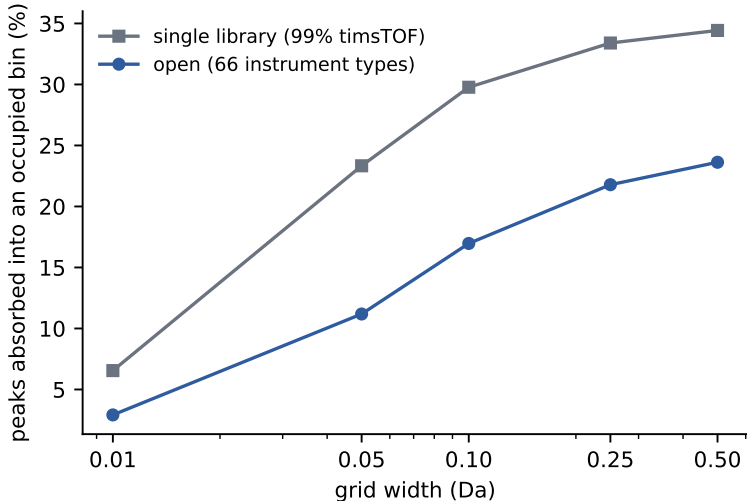


Figure 4: Fraction of peaks absorbed into an already-occupied bin against grid width, on each corpus’s own held-out spectra. This is a descriptive property of the representation, computed before any model is fitted; it is not evidence that absorption causes the performance differences reported above.

10. The bin-width sweep remains exploratory

On the sensitivity corpus, the single-seed sweep gives macro AUPRC values of 0.0867 at 0.5 Da, 0.0818 at 0.25 Da, 0.0801 at 0.1 Da, 0.0909 at 0.05 Da, 0.1030 at 0.01 Da and 0.1119 at full precision

(Table 5). The non-monotonic values at 0.25 and 0.1 Da are compatible with training noise in a single run and should not be read as evidence that coarser grids are better. The 0.01 Da model closes approximately 65% of the 0.5 Da-to-full-precision gap. Because each intermediate width was run once and on one corpus, the apparent transition below 0.05 Da is exploratory rather than a transferable threshold.

Table 5: Single-seed bin-width sweep on the sensitivity corpus. Every arm uses the M1 encoder at an identical parameter count; only the width of the mass grid changes. The endpoints (0.5 Da and full precision) correspond to M1R and M1 and are supported by three seeds elsewhere in this work; the four intermediate widths are single exploratory runs.

Bin width (Da)	Status	Macro AUPRC	Share of 0.5 Da gap closed
0.5	endpoint (M1R)	0.0867	0%
0.25	single-seed exploratory	0.0818	-19%
0.1	single-seed exploratory	0.0801	-26%
0.05	single-seed exploratory	0.0909	17%
0.01	single-seed exploratory	0.1030	65%
Full precision	endpoint (M1)	0.1119	100%

11. Target and scaffold resampling quantify different sources of uncertainty

The target-resampling intervals above hold the held-out set fixed. The scaffold-cluster bootstrap resamples it: 2,000 draws over the 3,784 scaffold clusters partitioning the 10,058 held-out structures, recomputing macro AUPRC per arm and forming the contrasts within each draw.

Table 6 gives the result. The estimator was verified before use: run against the full held-out set it reproduces all seven reported point estimates exactly, which is the check that the resample estimates the same quantity as the reported metric. Because average precision is nonlinear under duplication, the bootstrap mean is not expected to coincide with the point estimate, and the two are shown together rather than one standing for the other.

Table 6: Scaffold-cluster bootstrap on the primary corpus, against the untuned dense baseline: 2,000 draws over the 3,784 scaffold clusters partitioning the 10,058 held-out structures, with macro AUPRC recomputed per arm and contrasts formed within each draw. The bootstrap mean is shown beside the point estimate so that centring can be judged directly. The two contrasts whose arms share the peak encoder and the peak-pooling rule centre to within 0.0009; every contrast whose arms differ in peak pooling or in network family carries an upward bias of +0.0026 to +0.0073, and those intervals should be read as locating a range rather than as centred on the estimate.

Contrast	Kind	Point estimate	Bootstrap mean	SD	95% interval
M1D – M0	composite	+0.0172	+0.0198	0.0057	[+0.0123, +0.0317]
M1R – M1D	composite	−0.0023	+0.0011	0.0048	[−0.0051, +0.0105]
M1R – M0	composite	+0.0149	+0.0209	0.0096	[+0.0089, +0.0377]
M1 – M1R	one-factor	+0.0142	+0.0151	0.0028	[+0.0108, +0.0216]
M2 – M1	composite	+0.0179	+0.0183	0.0026	[+0.0142, +0.0242]
M1 – M0	composite	+0.0291	+0.0360	0.0116	[+0.0213, +0.0557]
M2 – M0	composite	+0.0470	+0.0543	0.0133	[+0.0370, +0.0765]

Two features of this table bear on how the paper’s claims should be read. First, the two contrasts measured inside a fixed peak encoder and a fixed peak-pooling rule — the mass-axis intervention and the measurement model — are well behaved under resampling of structures: their

bootstrap means sit within 0.0009 and 0.0004 of their point estimates, their standard deviations are the smallest in the table, and their intervals exclude zero comfortably. Second, every contrast whose arms differ in peak pooling or in network family is biased upward, and by more the further apart the arms are — +0.0034 for M1R — M1D, +0.0026 for M1D — M0, +0.0060 for M1R — M0 and +0.0073 for M2 — M0. Resampling with replacement duplicates structures, which sharpens macro AUPRC; that sharpening cancels only between arms that rank molecules similarly. Contrasts across architectures are therefore less stable under this resampling than contrasts inside one, which is consistent with the reading given to them throughout.

The one interval that changes an interpretation is M1R — M1D. Its target-resampling interval, $[-0.0066, +0.0020]$, contains zero; its scaffold-cluster interval, $[-0.0051, +0.0105]$, also contains zero but is displaced upward. Neither supports a claim in either direction, which is the conclusion already drawn in Sect. 6 for independent reasons.

These intervals are wider than the target-resampling intervals, as expected: they carry variability from which structures were held out, which the target bootstrap holds fixed by construction. Neither is a confidence interval for the population of MS/MS tasks. Reported together with the seed standard deviations, they say that the mass-axis intervention is reproducible across all three sources, and that the ordering of the mass-axis and token-encoder-package increments is not.

Discussion

The central result is not that a binned representation outperforms peak tokens. It is that the apparent cross-architecture ordering is not robust to comparator construction. A two-arm comparison between the original unconditioned binned baseline and the full-precision peak-token model yields +0.0579 macro AUPRC. Two interventions on the comparator alone, neither touching the treatment, remove it. Supplying the dense baseline with the acquisition covariates the peak-token models already had accounts for +0.0288. Giving it its own hyperparameter search accounts for +0.1183 more, and reverses the ordering: every peak-token arm then sits below the baseline in every seed. The same grid run on the peak-token side does not restore it, being worth +0.0006 there, so the reversal is not an artefact of searching one arm only.

The sequence is worth stating as a chain, because each step licenses a different statement. In the original comparison the peak-token pipeline appears better. Matching acquisition information substantially reduces the gap. Architecture-specific optimisation of the comparator reverses the ordering. Matched search effort on the peak-token arm does not restore it. The capacity mismatch that remains prevents a representation-level winner from being declared. Independently of all four steps, quantising mass within a fixed peak-token architecture isolates a reproducible contribution of full mass precision.

Neither comparator intervention is exotic. Metadata parity is a matter of passing the same inputs to both sides. A per-arm hyperparameter search is standard practice that was not applied in the original design, and its omission was not neutral, because the shared schedule and the comparator’s width both came from the treatment arm. Running the same search budget on both sides shows how unequal that inheritance was: the learning rate the comparator selects is the worst region of the grid for the treatment, and the width each selects sits at the opposite end from the other’s. This result indicates that a study can satisfy the usual expectations — identical data, identical splits, identical objective, identical training budget, three seeds — and still report an ordering that inverts under a control it did not run.

Within the peak-token architecture, the rounding control gives the one contrast in which a single component moves: restoring full mass precision contributes +0.0142, measured with the encoder, pooling, parameter count and schedule held fixed. Mass precision is therefore not unimportant — it carries real predictive information, measured here without architectural confounding, and it is the most reproducible effect in the study across seeds, targets and structures alike.

The remaining +0.0149 should not be described as an encoder effect. M1R and M0 differ in input interface, network family, peak pooling and conditioning mechanism simultaneously. The robust conclusion is therefore not that encoder and mass axis are equal, but that the apparent full-precision advantage is not dominated by mass resolution once the comparator receives the same acquisition variables.

The attention-free sensitivity arm supports a similarly narrow reading. M1R does not outperform M1D, but the arms differ in pooling and feed-forward width in addition to the attention operator. The experiment shows that the tested Transformer-style attention-bearing configuration is not necessary to obtain the rounded-token performance observed here; it does not identify the isolated effect of self-attention, and we have specified the ablation that would.

These distinctions matter because the purpose of the study is attribution. When representation, interface, architecture, pooling and metadata availability move together, a performance increment belongs to the whole package unless the design contains an intervention that changes only one component. The rounding control satisfies that requirement for mass-axis precision, and it is the only contrast here that does. The measurement-model arm does not: part of its increment appears on molecules represented by a single spectrum, where there is nothing to aggregate, so it bundles a change of output and evidence model with the change of pooling rule. The metadata correction establishes a separate and practically important point: comparator *inputs* can account for a performance difference as large as the modelling changes being studied.

Relation to other work

The systems this work sits alongside are not benchmarked here, and could not be without a different study design. CSI:FingerID [3], MIST [4], DreaMS [6], MS2DeepScore [5] and TransExION [11] differ from this work and from one another in training corpus, prediction target, objective and architecture; a macro AUPRC over BRICS fragments is not comparable to a retrieval rank or a similarity score. Their datasets, prediction targets, evaluation metrics, architectures and objectives all differ from ours and from one another, and this study makes no claim to outperform any of them. It is an attribution and control study, not a leaderboard entry, and it does not establish why any of those systems outperforms its own baseline.

What it offers those systems is a control strategy and an empirical warning. Featurisation benchmarks such as Giné et al. [7] measure each representation bundled with the encoder that reads it; that is the right design for choosing a pipeline and the wrong one for attributing a gain, and the rounding control is a cheap way to add the missing arm. Benchmarks such as MassSpecGym [9] have standardised splits and leakage control, which is a different axis from attribution and complementary to it: a leakage-controlled split makes a number trustworthy, while metadata parity and a within-encoder control make it interpretable. For metadata-conditioned models specifically, our result implies that reporting what the comparator was told is as important as reporting what the model was told. For multi-spectrum aggregation, the +0.0179 contributed by the measurement model here is a reminder that a component unrelated to spectral representation can produce an effect of the same size as the representation change under study, and should not be left uncontrolled in a comparison whose conclusion concerns representation.

Interpretation of the sensitivity corpus

The contrasting corpus changes the acquisition regime substantially, which is why it is informative, but it is not an independent replication. The larger mass-axis increment on the nearly all-timsTOF corpus is compatible with the hypothesis that preserving reported mass precision is more valuable when the instrument genuinely supplies high-resolution measurements. That explanation is plausible and untested here; the appropriate next analysis stratifies the M1-minus-M1R effect by instrument-resolution class using the saved held-out predictions, which requires no retraining.

The bin-width sweep is likewise suggestive rather than definitive. It indicates that a 0.01 Da grid recovers much of, but not all of, the full-precision advantage on the sensitivity corpus. This differs below 0.1 Da from the result reported by de Jonge et al. [8]. The tasks, architectures and training stochasticity differ, and the present sweep has one seed at each intermediate width, so neither study alone settles a universal sufficient mass resolution.

Limitations

A. Search and capacity. The two searches equalise search effort over architecture-specific capacity parameters; they do not equalise parameter count or effective capacity. The tuned dense baseline carries 4,442,128 parameters against the tuned peak-token arm’s 712,336, a factor of 6.2, and the two grids themselves span different capacity ranges. The grid is modest, six learning-rate-by-width combinations at two dropout rates, and the dense baseline’s selection lies on its boundary in both axes, which indicates that the search may not have located that architecture’s optimum; the direction of any change under a broader search cannot be assumed. The peak-token arm’s search selected the configuration it already had, which indicates that the arm was near its optimum on this grid but not that a wider grid would leave it there. Only M1R was searched on the peak-token side; M1, M1D and M2 inherit its configuration.

B. Statistical uncertainty. Three seeds per arm are sufficient to show the observed seed-to-seed pattern but not to establish rankings among small effects. The target-resampling bootstrap does not capture uncertainty in which structures were held out; the scaffold-cluster bootstrap addresses that but holds the target set fixed. BRICS targets are correlated because fragments co-occur, so target-resampling intervals are narrower than fully independent resampling would give. Average precision is nonlinear under duplication, so bootstrap means need not coincide with point estimates, and the percentile intervals reported here are not bias corrected. A scaffold-cluster jackknife or a weighted bootstrap would test how far the intervals depend on that choice; neither was run. Only one scaffold partition was used, so no interval here covers the choice of fold assignment, and none covers the population of MS/MS tasks.

C. Ablation purity. Only M1-to-M1R changes one component. M1D-to-M1R is not a pure attention contrast, because the peak-pooling rule and the feed-forward width change with the operator; it is reported as a sensitivity analysis. M1R-to-M0 is composite in four respects at once, including the mechanism by which metadata are injected; a dense baseline conditioned through adaptive normalisation rather than concatenation would test whether the +0.0149 depends on that mechanism. M2-to-M1 is composite: the measurement model changes the output and evidence model as well as the pooling rule, and a substantial part of its increment is present on molecules with a single spectrum. Evaluation also uses at most four spectra per molecule, although 2,755 of the 10,058 held-out molecules have more, so the reported figure understates what pooling could contribute under a larger cap; the claim in Methods that the additive sufficient statistics impose no cap on K_i describes the estimator rather than the experiment run here.

D. Data. The sensitivity corpus is a deliberately contrasting in-house slice of the same aggregate

resource, processed by the same code. It is neither external validation nor independent replication, and it is not redistributable, so that arm of the study cannot be reproduced by a third party from public data. The comparator searches were run on the primary corpus only, so the reversal reported in Sects. 2 and 3 is untested on the sensitivity corpus. The instrument-resolution explanation for the difference between corpora is not tested directly. The primary corpus is assembled from public sources and its selection is scripted, but the aggregated file carries no per-record licence field, so redistributability cannot be established record by record.

E. Endpoint. The endpoint is BRICS substructure prediction over 512 frequent fragments. The study makes no claim about candidate ranking, full molecular fingerprints or de-novo structure elucidation, and transfer to those endpoints remains to be demonstrated.

Conclusion

This study reports conclusions at three levels.

Comparator robustness. The apparent advantage of a full-precision peak-token model over a binned baseline was not robust to comparator construction. On the primary corpus an unconditioned 0.5 Da binned baseline reaches 0.1130 macro AUPRC against the peak-token model’s 0.1709. Supplying the baseline with collision energy, adduct, polarity and instrument raises it to 0.1418. An architecture-specific search of its own raises it to 0.2601 ± 0.0008 and reverses the ordering. The same search budget applied to the peak-token arm’s own capacity parameters raises it from 0.1567 to 0.1573 ± 0.0036 and selects the configuration that arm already had, leaving the tuned-against-tuned difference at -0.1028 . Because the optimised dense baseline carries $6.2\times$ the parameters of the optimised peak-token arm, this reversal does not establish that binned representations are superior.

Controlled representation effect. Within a fixed peak-token architecture, restoring full mass precision contributes $+0.0142$ macro AUPRC ($+0.0097$ to $+0.0187$), with the encoder, pooling rule, parameter count and schedule held fixed. This is the study’s only single-factor intervention and its most reproducible effect across seeds, targets and held-out structures. The statement stands independently of how the binned baseline is built, because no binned baseline enters it. The measurement model contributes $+0.0179$ on the same comparator-free footing, but that figure bundles multi-spectrum pooling with a change of output and evidence model and is reported as a composite contrast.

Methodological recommendation. Cross-architecture representation studies should supply identical auxiliary information to every arm; give each architecture its own reasonable optimisation over its own capacity parameters; report parameter counts and capacity differences explicitly; and use within-architecture controls when making causal claims about representation. In this study the comparator corrections were larger than the modelling effects they were meant to contextualise: $+0.0288$ for metadata parity and $+0.1183$ for optimisation, against $+0.0142$ for the mass axis and $+0.0179$ for the measurement model. The comparator’s capacity was moreover inherited from the treatment, its hidden width set to twice the peak-token encoder’s model dimension and its schedule chosen for the treatment’s optimisation behaviour, so the asymmetry was present in the design before any data was seen.

These conclusions concern experimental attribution rather than a universal ranking of binned and peak-level MS/MS representations, and they are specific to this endpoint, these corpora and these architectures.

Supplementary Information

Additional file 1 reports per-arm and per-seed metrics for the 27 ladder runs and the six runs of the two searched arms, the full selection tables for both searches, the per-seed effect sizes and within-seed ordering checks, the bootstrap distributions, the peak-absorption analysis by adduct, collision energy and instrument family, the bin-width sweep, the split-integrity audit, and the commands that regenerate every analysis. Section S8 states both controls as compact algorithms so that the method can be applied without the repository.

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Author contributions

X.P. designed the study, implemented the models and controls, performed the experiments and analyses, and wrote the manuscript. C.W. contributed to interpretation of the chemical results and revised the manuscript. Both authors read and approved the final manuscript.

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Availability of data and materials

Code. All code, configurations, trained-run reports and analysis scripts are available at <https://github.com/xulinpan/cheminformatics> under the MIT licence. The two controls are additionally released as `msms-controls`, a NumPy-only library under the MIT licence, so that the rounding control and the conditioning-parity check can be applied to other models without installing this project’s training stack. Results in this paper correspond to a tagged release of that repository. A DOI-bearing archive of that release has not yet been deposited, so no persistent identifier is quoted here. [OUTSTANDING: DEPOSIT THE ARCHIVE AND QUOTE ITS DOI]

Environment. `requirements-lock.txt` pins every dependency at the versions used for the reported runs (PyTorch 2.11.0+cu128, Python 3.14, NumPy 2.4.4, pandas 2.3.3, RDKit 2026.3.6, pyarrow 24.0.0), and `pyproject.toml` declares the supported ranges. Continuous integration runs the test suites on Python 3.10–3.13.

Primary corpus. The public-source primary corpus consists of the 414,992 spectra whose `ingest_lib` field names GNPS (220,849), MassBank (101,727) or MoNA (92,416), selected by `scripts/make_open_subset.py` from the aggregated CASMI 2026 file distributed by the competition organisers. Two limitations of that route must be stated rather than implied. The aggregated file carries no per-record licence column, and all three sources declare licences per record, so library membership does not establish that any individual record is redistributable. It also carries no accession, spectrum identifier or USI column, so the selected records cannot be matched back to their source entries or re-fetched from GNPS, MassBank and MoNA by identifier. Reproducing this corpus therefore currently requires the aggregated file, and assembling a

redistributable equivalent would require re-deriving it from the sources with those fields retained. `docs/corpus_composition.md` records each source library’s contribution and which parts are redistributable.

Filtering and identity. Records are retained if precursor $m/z < 2,000$ and reported precursor error is within 20 ppm; peaks above 0.1% of the base peak and below precursor +1.5 Da are kept, capped at the 256 most intense. Molecular identity is the 14-character InChIKey first block, and folds are assigned by generic Murcko scaffold group as described in Methods; `scripts/split_integrity_audit.py` reproduces the audit reported there.

Artefacts. The repository contains the per-seed metrics for every run, the per-target average precision for every arm and seed, the bootstrap draws, the stratified absorption counts, the trained-run configurations, and the figure- and table-generating scripts, under `results/` and `scripts/`.

Sensitivity corpus. The in-house libraries in the CASMI aggregate are not redistributable and are not used in the primary experiments. The sensitivity corpus drawn from them cannot be reproduced by a third party, and no conclusion in this paper rests on it alone.

Declarations

Competing interests

The authors declare no competing interests.

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